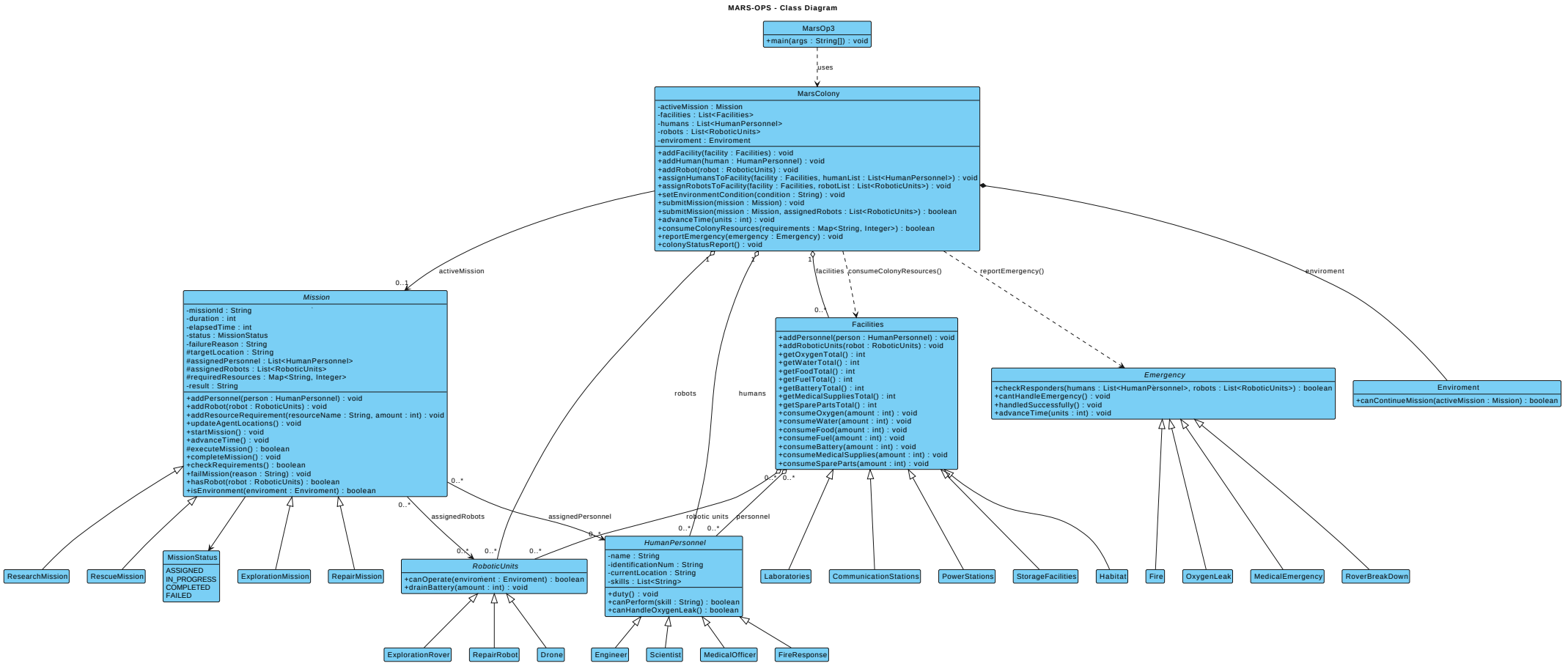

MARS-OPS - Class Diagram
MarsOp3
+main(args : String[]) : void
uses
MarsColony
-activeMission : Mission
-facilities : List<Facilities>
-humans : List<HumanPersonnel>
-robots : List<RoboticUnits>
-enviroment : Enviroment
+addFacility(facility : Facilities) : void
+addHuman(human : HumanPersonnel) : void
+addRobot(robot : RoboticUnits) : void
+assignHumansToFacility(facility : Facilities, humanList : List<HumanPersonnel>) : void
+assignRobotsToFacility(facility : Facilities, robotList : List<RoboticUnits>) : void
+setEnvironmentCondition(condition : String) : void
+submitMission(mission : Mission) : void
+submitMission(mission : Mission, assignedRobots : List<RoboticUnits>) : boolean
+advanceTime(units : int) : void
+consumeColonyResources(requirements : Map<String, Integer>) : boolean
+reportEmergency(emergency : Emergency) : void
+colonyStatusReport() : void
activeMission
0..1
1
facilities
consumeColonyResources()
reportEmergency()
enviroment
Mission
-missionId : String
-duration : int
-elapsedTime : int
-status : MissionStatus
-failureReason : String
#targetLocation : String
#assignedPersonnel : List<HumanPersonnel>
#assignedRobots : List<RoboticUnits>
#requiredResources : Map<String, Integer>
-result : String
+addPersonnel(person : HumanPersonnel) : void
+addRobot(robot : RoboticUnits) : void
+addResourceRequirement(resourceName : String, amount : int) : void
+updateAgentLocations() : void
+startMission() : void
+advanceTime() : void
#executeMission() : boolean
+completeMission() : void
+checkRequirements() : boolean
+failMission(reason : String) : void
+hasRobot(robot : RoboticUnits) : boolean
+isEnvironment(enviroment : Enviroment) : boolean
Facilities
+addPersonnel(person : HumanPersonnel) : void
+addRoboticUnits(robot : RoboticUnits) : void
+getOxygenTotal() : int
+getWaterTotal() : int
+getFoodTotal() : int
+getFuelTotal() : int
+getBatteryTotal() : int
+getMedicalSuppliesTotal() : int
+getSparePartsTotal() : int
+consumeOxygen(amount : int) : void
+consumeWater(amount : int) : void
+consumeFood(amount : int) : void
+consumeFuel(amount : int) : void
+consumeBattery(amount : int) : void
+consumeMedicalSupplies(amount : int) : void
+consumeSpareParts(amount : int) : void
Emergency
+checkResponders(humans : List<HumanPersonnel>, robots : List<RoboticUnits>) : boolean
+cantHandleEmergency() : void
+handledSuccessfully() : void
+advanceTime(units : int) : void
Enviroment
+canContinueMission(activeMission : Mission) : boolean
robots
humans
0..*
assignedRobots
assignedPersonnel
robotic units
personnel
MissionStatus
ASSIGNED
IN_PROGRESS
COMPLETED
FAILED
ResearchMission
RescueMission
ExplorationMission
RepairMission
RoboticUnits
+canOperate(enviroment : Enviroment) : boolean
+drainBattery(amount : int) : void
HumanPersonnel
-name : String
-identificationNum : String
-currentLocation : String
-skills : List<String>
+duty() : void
+canPerform(skill : String) : boolean
+canHandleOxygenLeak() : boolean
Laboratories
CommunicationStations
PowerStations
StorageFacilities
Habitat
Fire
OxygenLeak
MedicalEmergency
RoverBreakDown
ExplorationRover
RepairRobot
Drone
Engineer
Scientist
MedicalOfficer
FireResponse